



TO STAY OR TO GO: PREDICTING CUSTOMER CHURN

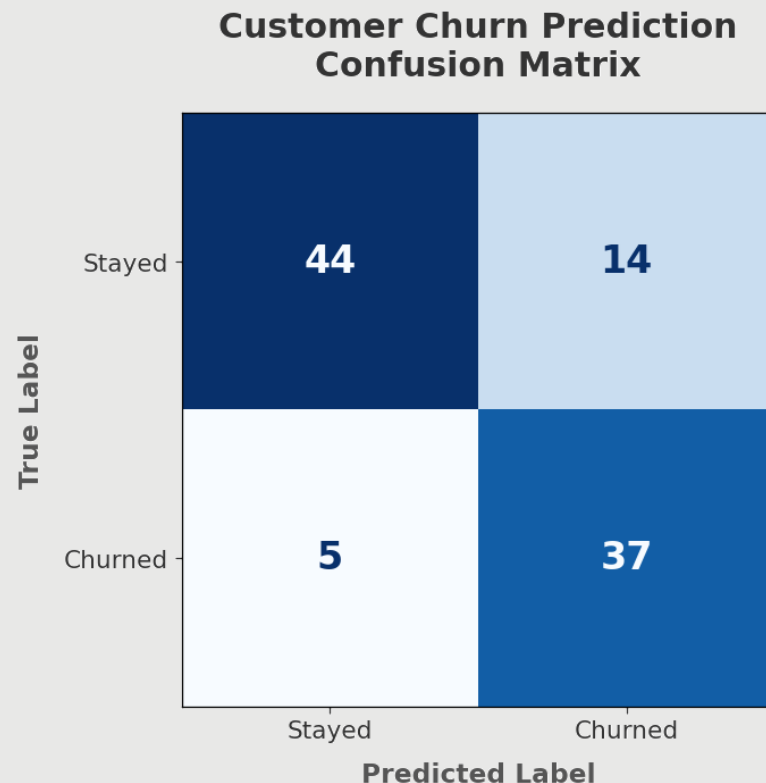
Charter Data Scientist Interview

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- Data
 - *Accounts (price and churn status)*
 - *Streams (timing and quality)*
- Customer Churn
 - *41% overall*
 - 63% churn for inactive users
 - 23% churn for active users
- Feature Engineering
 - *Volume*
 - *Recency*
 - *Quality*
 - *Value*
- Modeling Considerations
 - *Binary Classification*
 - *Structural Missingness in Data*

MODEL AND PREDICTION PERFORMANCE



- XGBoost Classifier
 - *Data Split: 80%/10%/10%*
 - Stratified by churn status
 - *Objective: Precision-Recall AUC*
 - *Randomized Hyperparameter Tuning*
 - *Validation Based Early Stopping*
- Performance
 - *PR-AUC: 0.75*
 - *F1-Score: 0.80*
 - *Precision: 0.73*
 - *Recall: 0.88*
 - *Accuracy: 0.81*
- Feature Importance
 - *Recency >>> Volume Quality Value*
- Future Work
 - *Build separate models based on activity level*
 - *Measure changes in user behavior over time*

AI USAGE AND IMPLEMENTATION



- AI Usage
 - *Coding*
 - Double checking implementation
 - Troubleshooting errors
 - *Modeling*
 - Feature suggestion
 - Initial value selection
 - *Communication*
 - Presentation outline
 - Refining visualizations
- Implemented in Python
 - *Polars*
 - *XGBoost*
 - *scikit-learn*
- github.com/nageljo/CharterInterview